THE FUNDAMENTAL COMPUTATION OF THE CLASS OF THE VIRTUAL NORMAL SHEAF AGAINST THE FUNDAMENTAL CLASS OF A REGULARLY EMBEDDED SUBVARIETY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBM (sources/stacks-chow/pinned-original-source.tex line 12076).

Allowed standard prerequisites (editorial).

- The class of the virtual normal sheaf and its base change behaviour. Same chapter, Lemma labelled lemma-construction-gysin, cited at proof lines 12087-12089. Allows X to be replaced by Y in the computation.
- Decomposition of the class of the virtual normal sheaf. Same chapter, Lemma labelled lemma-gysin-decompose, cited at proof lines 12097-12099. Reduces to the case where the normal and conormal sheaves are identified.
- The class C at infinity and the Gysin class of the divisor W_infinity. Same chapter, Lemma labelled lemma-gysin-at-infty, invoked at proof lines 12106-12112, and Lemma labelled lemma-easy-gysin, cited at proof lines 12116-12118. Computes C cap [X]_n as [W_infinity]_n.
- The symmetric algebra structure on the conormal sheaf of a regular immersion. Divisors, Lemma labelled divisors-lemma-quasi-regular-immersion, cited at proof lines 12120-12122. Identifies the normal cone with the vector bundle of the conormal sheaf, giving p^*[Z]_{n - r} = [C_Z X]_n.

Source reference. Original reference 0FA4: sources/stacks-chow/pinned-original-source.tex, source lines 7677–7677. Same-chapter pointer into the author environment remark at source line 7677; invoked at proof line 12089 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBK: sources/stacks-chow/pinned-original-source.tex, source lines 11912–11912. Same-chapter pointer into the author environment lemma at source line 11912; invoked at proof line 12090 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0FBL: sources/stacks-chow/pinned-original-source.tex, source lines 11994–11994. Same-chapter pointer into the author environment lemma at source line 11994; invoked at proof line 12095 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line 12106 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02TB: sources/stacks-chow/pinned-original-source.tex, source lines 5170–5170. Same-chapter pointer into the author environment lemma at source line 5170; invoked at proof line 12112 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 063M: sources/stacks-chow/tags.txt, source lines 7836–7836. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12120 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In the situation described just above assume* $\dim_\delta(Y) = n$ *and that* $\mathcal{C}_{Y \times_X Z/Z}$ *has constant rank* r*. Then*

$$c(Z \to X, \mathcal{N}) \cap [Y]_n = c_{top}(\mathcal{E}) \cap [Z \times_X Y]_{n-r}$$

in $\mathrm{CH}_*(Z \times_X Y)$*.*

Proof. The bivariant class $c_{top}(\mathcal{E}) \in A^*(Z \times_X Y)$ was defined in Remark 0FA4. By Lemma 0FBK we may replace X by Y. Thus we may assume $Z \to X$ is a regular closed immersion of codimension r, we have $\dim_\delta(X) = n$, and we have to show that $c(Z \to X, \mathcal{N}) \cap [X]_n = c_{top}(\mathcal{E}) \cap [Z]_{n-r}$ in $\mathrm{CH}_*(Z)$. By Lemma 0FBL we may even assume $\mathcal{N}^\vee \to \mathcal{C}_{Z/X}$ is an isomorphism. In other words, we have to show $c(Z \to X, \mathcal{C}_{Z/X}^\vee) \cap [X]_n = [Z]_{n-r}$ in $\mathrm{CH}_*(Z)$.

Let us trace through the steps in the definition of $c(Z \to X, \mathcal{C}_{Z/X}^\vee) \cap [X]_n$. Let $b : W \to \mathbf{P}^1_X$ be the blowing up of $\infty(Z)$. We first have to compute $C \cap [X]_n$ where $C \in A^0(W_\infty \to X)$ is the class of Lemma 0F9J. To do this, note that $[W]_{n+1}$ is a cycle on W whose restriction to $\mathbf{A}^1_X$ is equal to the flat pullback of $[X]_n$. Hence $C \cap [X]_n$ is equal to $i_\infty^*[W]_{n+1}$. Since W_∞ is an effective Cartier divisor on W we have $i_\infty^*[W]_{n+1} = [W_\infty]_n$, see Lemma 02TB. The restriction of this class to the open $C_Z X \subset W_\infty$ is of course just $[C_Z X]_n$. Because $Z \subset X$ is regularly embedded we have

$$\mathcal{C}_{Z/X,*} = \mathrm{Sym}(\mathcal{C}_{Z/X})$$

as graded $\mathcal{O}_Z$-algebras, see Divisors, Lemma 063M. Hence $p : N = C_Z X \to Z$ is the structure morphism of the vector bundle associated to the finite locally free module $\mathcal{C}_{Z/X}$ of rank r. Then it is clear that $p^*[Z]_{n-r} = [C_Z X]_n$ and the proof is complete. □